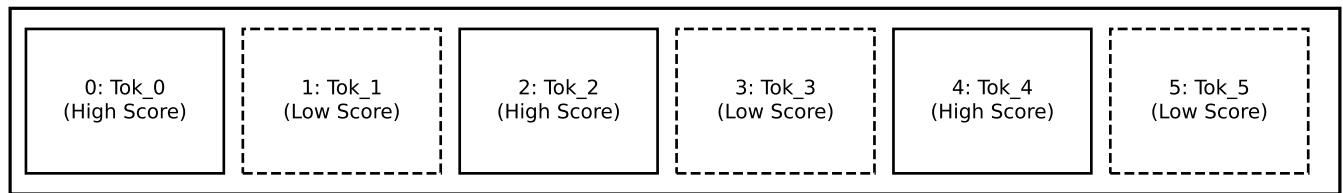


MIXTURE-OF-EXPERTS (MoE) TOKEN BUFFER STATE TRANSFORMATION

STATE A: Candidate Expert Token Buffer (Before Compaction)



IDEMPOTENT CAPACITY PROJECTION & DISJOINT CYCLE RESOLUTION

Idempotent Condition: $f(f(x)) = f(x)$ over Expert Token Space

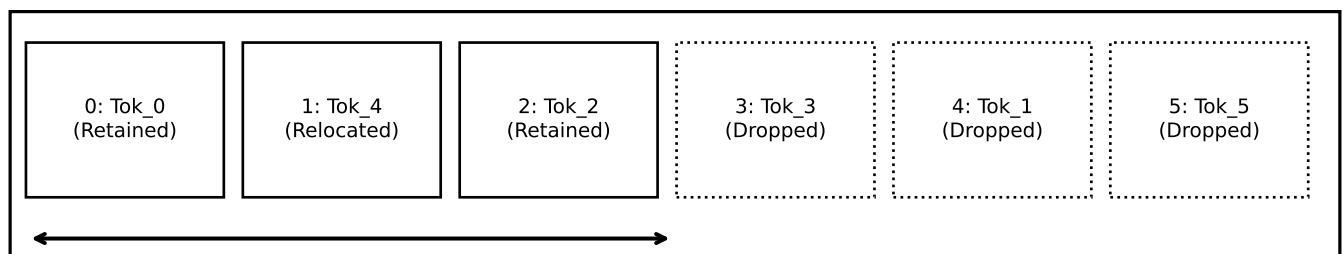
- Fixed Points ($f(i) = i$): Slots 0, 2 are high-affinity tokens already within capacity C.

- 2-Cycle Transposition Orbit (Slot 1 $\leftrightarrow$ Slot 4):

* Slot 1 (Low Score in Active Zone) swapped with Slot 4 (High Score in Tail Zone)

* $O(1)$ In-Register Vectorized Exchange along Hidden Dimension $D = 1024$

STATE B: Compacted Active Expert Buffer (After Capacity Truncation)



Contiguous Active Expert Tokens (C=3)
Ready for High-Throughput GEMM on Tensor Cores

FIG. 2